

Scopus SoW PoC

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1. Overview

The goal is to develop a device for detecting objects using a camera, capture photos, save them to SD card and upload to a remote server over cellular network. Additionally, to notify remote server of detected activity.

For a PoC phase the device is controlled locally by shell commands.

The base points are existing N6 Camera application (N6) and WP76 Modem (Mangoh) board running SDVR application.

2. Block Diagram

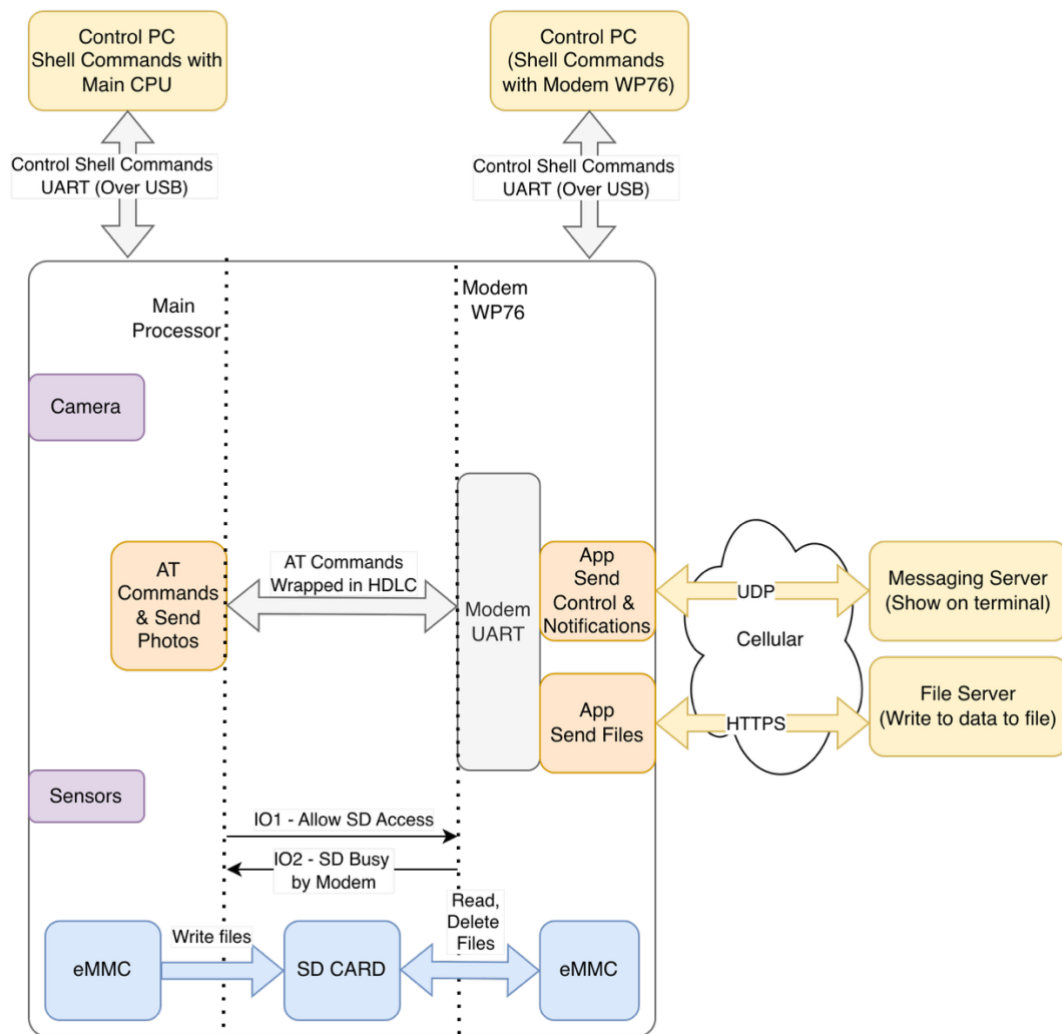


Figure 1: Conceptual Block Diagram

3. Functions

3.1 Operation Commands (N6)

New Commands

- Start/Stop objects detection and counting, defines what to detect people, cars or both.
- Enable/Disable objects notifications transmission to remote server.
- Enable/Disable taking photo and saving to SD card.
- Enable/Disable taking photo and send to remote server (by cellular modem).
- Enable/Disable taking photo and saving to SD card on detection of new objects.
- Enable/Disable taking photo and sending to remote server on detection of new objects.
- Query taking photo actions settings.

3.2 SD Card Management Commands (SDVR app)

Existing commands

These commands are tunneled to the modem, through main processor (N6).

- Mount/Unmount SD card
- Format SD card
- List files on SD card
- Command to upload all files to the server
- Command to upload a specific file to the server
- Delete all files from SD card
- Delete specific file from SD card.
- Configure Communication Settings from SD card configuration file.

3.3 Camera Configuration (N6)

Based on existing shell commands on the board.

3.4 Photo Configuration (N6)

JPEG file configuration, set and query current settings: Color space, Chroma subsampling, image Height, Width and Quality.

3.5 Sensors Configuration (N6)

- IR sensor operation, enable, disable and query settings.
- Motion sensor, set sensitivity, trigger, and query settings.

3.6 Cellular communication configuration (SDVR App)

These commands are tunneled to the modem, when received by the shell interface of N6, sent over another UART to the modem.

Existing Commands

- All standard AT commands supported by the modem
- Setup and query APN credentials
- Setup and query HTTP endpoint host (IP address or URL), port, path for uploading files.
- Import and query certificates.

New Command:

- Setup and query destination IP address and port for Notifications & Control Channel

3.7 System Commands (N6)

Existing commands:

- Query RTC time
- Query system version details of main CPU.
- List of supported shell commands including sub commands and parameters

New commands:

- Set RTC time (New)
- Query application version

3.8 Local Notifications shell (N6)

New notifications

All events are streamed to the shell interface port, including:

- Modem tunneled responses such as start, finish or error of file upload operation whether read from SD or directly from UART.
- Detection of new objects, from camera
- Photo taken event
- Motion start/stop notifications
- Responses for query of current settings.

4. Control Channel Communication (PC - Main CPU)

4.1 General Settings

- Textual channel, binary sent as hex – string
- Shell like commands, with sub commands and parameters with separator.
- Command ends with terminator char, CR.
- Echo can be turned on or off.
- Each command returns either result ok, in form of cmd [subcommand] ok or specific error related to the command.
- Commands are divided into groups with main and sub commands with or without parameters:
 - Main-command
 - Main-command parameter-1 parameter-2
 - Main-command sub-command
 - Main-command sub-command parameter-1 parameter-2
- Appended to existing CPU shell commands
- Bitmasks should be treated as unsigned int 32 bits, specified are bit values, for example for bit 0, value will be 1.
- Settings should be saved into non-volatile memory and read upon device reset or by demand.

4.2 Operation Commands (N6)

New shell commands, these processes can be run from a computer connected to a terminal.

Command	Description
detect start	Start object detection and counting process
detect stop	Stop detection
detect profile <det_msk> <action_msk>	Set profile of objects to detect <det_msk> bitmask values: 1 – for people, 2 – vehicles <action_msk> bitmask values: 1 – save to SD, 2 – Report by cellular
detect profile query	Query current profile, response: detect profile: <det_msk> <action_msk>
notify enable <notify_msk>	Enable notifications, according to bit mask Bit mask includes: x1 – Network registration x2 – Motion Start x4 – Motion Stop x8 – Periodic notifications x10 – People detected x20 – Vehicle detected .. TBD
notify disable	Disable notifications
notify trigger <notif_code>	Manually send notification. XXX – notification code
notify period <seconds>	Set interval in seconds for periodic notifications
notify query	Returns status of requested notifications
photo savesd	Trigger to take photo regardless of conditions and save it to SD card
photo upload	Trigger to take photo regardless of conditions and upload it by HTTPS
rtc set DDMMYYYYHHMMSS	Set system RTC clock. The clock should be updated automatically on modem network registration.

Notes

- Saving photo to SD card involves acquiring access to SD, if possible, if not response with error.

4.3 Camera Settings Commands (N6)

Keep existing shell commands, as available on the main CPU (flip, aec, awb, gain, exposure, brightness)

4.4 Saved Photo Settings Commands (N6)

Command	Description
img size <height> <width>	Set image size height and width in pixels
img quality <quality>	Set image quality (1 – 100)
img color <YCBCR/RGB/CMYK>	Color space
img chroma <profile>	Chroma subsampling profile <profile> 0 -> 4:4:4, 1 -> 4:2:2
img query	Returns photo settings in a row img: <height> <width> <quality> <color> <chroma>

4.5 Sensors Settings Commands (N6)

Command	Description
irled on	Turn IR Led on
irled off	Turn IR Led off
irled query	Query state of IR LEDs irled: 0/1
motion sense <sensitivity> <no motion timeout>	Set motion sensor sensitivity and timeout without motion to consider no motion (Halt) state.
motion query	Return current set parameters

4.6 SD Card and Cellular communication Commands (N6->SDVR)

- These commands are tunneled to the modem.
- Responses are tunneled directly according to modem protocol

Command	Description
mdm <modem or app command>	Tunnel commands to modem command for Cellular or SD communication

5. Modem Channel Communication (N6 - Modem)

5.1 AT Commands Interface

- AT interface, sent commands terminating with CR
- Responses are enclosed in <CRLF>OK<CRLF>
- All AT modem commands from the modem are passthrough (in N6) both directions including responses, URCs and errors (including quotes chars handling)
- Custom application extension commands, in addition to original modem commands.
- Two operation modes AT command (ASCII) used for short AT commands or short binary messages such as hex-string coded notifications.
- System error codes can be used as error codes in the respective response of the command.
- For direct CPU to modem communication, use HDLC protocol to wrap the messages, such that there won't be difference if the command is ASCII or binary.
- Existing and New commands notes are according to existing SDVR app.

5.2 Commands to send to the Modem from N6

Command	Description
Existing commands	
SDVR+CNF	Read the configuration file and perform commands
SDVR+CERTDEL="[filename]"	Delete the internal certificate file, default if filename is not given
SDVR+UPLSTOP	Stop uploading
SDVR+UPLALL	Start uploading all files
SDVR+UPL=X	X – number of recent files to upload (recent by file date, descending order)
SDVR+HOSTIP=[IP ADDR]	Server IP address
SDVR+HOST="URL"	Server Host name (URL), up to 63 characters
SDVR+PORT=XXXXX	Server port to use, up to 65535
SDVR+SRVRPATH="path"	Server path (up to 31 characters)
SDVR+PROGRINTR=X	X – number of seconds to produce the update, range: 0-999 seconds. If X = 0, stop producing the updates, until new command
SDVR+PING=XXX	Ping communication with SDVR application X – numerator from 0 – 255.
SDVR+LSNOUPL	Send list of all non-uploaded files
SDVR+LSALL	Send list of all files (including uploaded, not uploaded)
SDVR+RMUPL	Erase all uploaded files
SDVR+RMALL	Erase all files (uploaded or not uploaded, excluding configuration files)
SDVR+FMTSD	Format SD card
SDVR+MOUNTSD	Mount SD
SDVR+UNMOUNTSD	Unmount SD
SDVR+APPSTOP	Stop running application, gracefully close all handles/threads and unmount SD
SDVR+TOK=ABCDEF12	Token to append to HTTP Header
New commands	
SDVR+NTFA=N,SIZE,"message"	Transmit notification over notification channels (UDP) X - numerator SIZE – message size in bytes.
SDVR+NTFHOST=[IP ADDR]	Server IP address, for notifications reporting
SDVR+NTFPORT=XXXX	Port for notifications reporting
SDVR+SEENDBIN=X,"TAG","TIME",REF,SIZE	Start binary transmission, X numerator, to be returned with response TAG – associated tag with meta data TIME – timestamp in DDMMYYYYHHMMSS format REF – Numerator reference SIZE – size of entire data to be transmitted in bytes.

5.3 Modem responses

Result / URC	Description
Existing commands	Applies to both when uploading from SD card or from memory
+SDVRUPL: START,"filename"	When upload of a file starts specifying filename
+SDVRUPL: END,"filename"	End of file upload, specified by filename
+SDVRUPL: PROGR,"filename",1	Progress timeout, every X seconds
+SDVRUPL: PROGR,"filename",0	As above, for out of sync cases
+SDVRUPL: ERROR CODE	Error while uploading file, propagate error code from the API handling the upload communication, where possible, custom code for file not found or no HTTP Server communication failures. ERROR CODE >=0
+SDVRUPL: START	Start uploading process
+SDVRUPL: END	All files successfully uploaded.
Existing commands	No changes
+SDVRCNF: START	Start reading configuration file
+SDVRCNF: END	Finished reading and executing configuration file tasks
+SDVRCNF: ERROR CODE	Custom code for failure reason. ERROR CODE >=0
+SDVRCERT: IMPORTOK	Certificate import finished successfully
+SDVRCERT: DELOK	Delete certificate from internal storage finished successfully
+SDVRCERT: DELSDOK	Delete certificate from SD card
+SDVRCERT: ERROR CODE	Numeric representation of error code
+SDVRPING: XXX	Return code for SDVR+PING command, XXX – received number
+SDVRRDY: version number	Fixed message to be send each time, the SDVR application starts With version code
+SDVRSRVR: OK	Response for successful server response (HTTP session finished)
+SDVRSRVR: ERROR CODE	Error returned by server
+SDVRDEL: PROGR,"filename",1	URC when deleting file in progress, every X seconds
+SDVRDEL: PROGR,"filename",0	As above, out of sync
+SDVRFMT: START	URC Start of format
+SDVRFMT: END	URC end of format
+SDVRFMT: PROGR,1	URC when format is in progress, produce every X seconds
+SDVRFMT: PROGR,0	URC when format is in progress, out of sync
+SDVRDEL: START	URC Start deleting files process
+SDVRDEL: END	URC End deleting files
+SDVRDEL: START,"filename"	URC before deleting file
+SDVRDEL: END,"filename"	URC after deleting file
+SDVRDEL: PROGR,"filename",1	URC when deleting file in progress, every X seconds
+SDVRDEL: PROGR,"filename",0	As above, for out of sync case

+SDVRBYE:	Finished all running all tasks, goodbye message, sent on application stop or modem turn off
New commands	
+SDVRNTF: END,X	Response for sending notification over UDP, return notification reference number X as received in request
+SDVRNTF: ERROR CODE	Error response for sending notification attempt to UDP, with error code, for example, error sending, no network, socket error, etc...
+SDVRNTF: "message"	Forward received message from remote UDP Server to main processor

6. Notifications & Control Messages Protocol (Modem - Remote Server UDP)

- UDP Client socket
- Up to 1400 bytes of payload
- JSON based payload
- More fields maybe added.
- Inbound tunnel commands – TBD
- Outbound tunnel responses – TBD
- Outbound message from Device to Remote Server

```
{
  // unit serial
  "ser": 12345,
  // Message numerator rollback at 0xFFFF
  "num": 10,
  // Transmission reason notification bitmask value such as periodic, motion, object detect, as decimal up to
  32 bits
  "rsn": 20,
  // Reason code additional data, according to message, ex: detected objects count
  "rsd": 4,
  // Timestamp of the event
  "tim": "YYYYMMDDHHMMSS",
  // Motion state 0 or 1
  "mtn": 0,
  // Additional information regarding operation mode
  "mod": "",
  "bat": 0.0,
  "vol": 0.0
}
```


7. Photos saving format Protocol (N6 to SD card)

New development

- File naming convention, including serial_DDMMYYYY_HHMMSS.rdy

Note: Currently file is saved as is without encryption.

8. Photos Upload to Remote Server (Modem – Remote HTTPS Server)

- All HTTPS communication is done by using certificates.

8.1 Upload of files from SD card

Extension of existing development

- Based on the existing SDVR, files to upload are saved with “rdy” extension. Change extension to “upd” after a successful upload.
- Custom HTTP Header to include **timestamp**, **ref**, **token** and **filename** as read from the SD card.
 - Timestamp is based on modem clock (or network).
 - Ref is a rollover incrementing index of the uploaded file in this session.

8.2 Live upload of files from UART

New development

- Custom HTTP headers to include **timestamp**, **ref**, **filename**, **token** as read from the TIME, REF, TAG parameters as received in AT command. Token is set separately by AT Command.
- Wait until all data, is received according to file size in AT command and stored in memory buffer.
- HTTP Transmission as binary, for example: Content-Type: “application/octet-stream”
- For In-Memory transmissions, wait until entire file is received and then upload.